

problem of losses during transport. There is a need to develop small-scale energy-storage units that could be located in people's homes, and give us an extremely robust, terrorist-resistant, delocalized electrical-energy system. Such an appliance might be developed using nanotechnology. With energy-storage units that are small, efficient and affordable, communities could buffer themselves against fluctuations in energy availability, such as when the sun stops shining or the wind stops blowing.

- 3. Efficient energy transmission** – In today's world, energy is not typically generated right where it is needed. Hence the need for energy transmission. Nanotechnology could be used to create new kinds of conductive materials that lose very little energy as electricity moves down the line. Many research groups are investigating whether nanowires and nanocoatings could reduce losses in electrical-transmission lines. "Self-cleaning" nanomaterials might be used to keep ice from accumulating on power lines—something that leads to disruptions in power in areas of cold winters. In the longer term, the need for efficient energy transmission might disappear if energy is converted and stored locally.
- 4. Efficient energy use** - Nanotechnology could lead to breakthroughs that indirectly conserve large amounts of fossil fuels. A large portion of the oil budget goes toward producing nitrogen fertilizers for agriculture. If scientists could find a way to fix nitrogen from the atmosphere, we could save huge amounts of fuel. Nanomanufacturing might also enable us to make all kinds of products using less energy. For instance, nanosensors might be used to track energy use and help minimize waste.



An Energized Future?

We only need to look around for ideas on harnessing energy.

These ideas will come from studies of living creatures—for instance, from learning about the way plants generate energy from sunlight and the way animals produce energy broadly throughout their bodies using their cells' mitochondria. Nanotools might help researchers create artificial life forms that resemble simple bacteria and then employ these creatures to produce fuel for.

Eventually, nanotechnology will lead to radical transformations in energy technologies. We only need to wait and see just how it will come about.